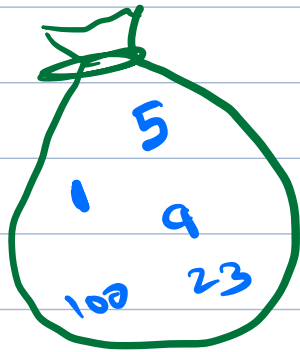


HashSet and HashMap

HashSet: collection of unique elements



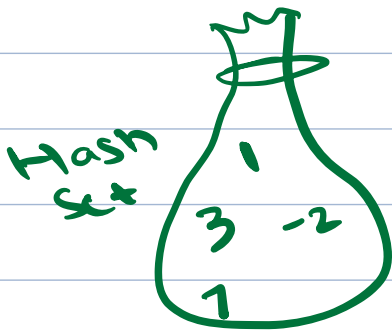
① Find element is present or not.

Examples which contain unique elements:

- ① Email addresses
- ② emp id / username
- ③ fingerprints

Array	Search n Linear Iteration	val at i^{th} idn $a[i]$
HS	Searching is faster	

arr = [1, 3, -2, 7, 1, 1, -2]



Hashset does not have any sequence of elements.

Syntax

```
HashSet <Type> hs = new HashSet <Type> ();
```

↳ wrapper class

int → Integer
long → Long

```
HashSet < Integer > hs = new HashSet < Integer > ();
```

Methods name. fn ()

① add (x) hs.add(5)

x gets added to hashset

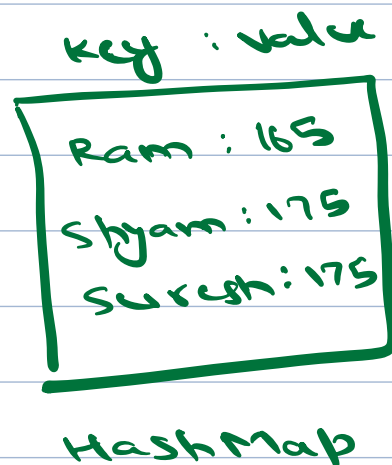
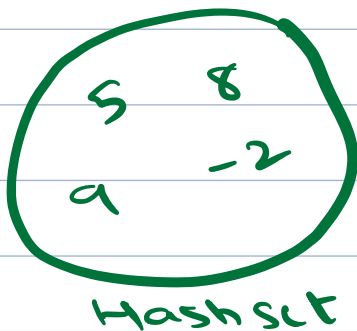
② remove (x) hs.remove(5)

x gets removed from hashset

③ size() hs.size()
no. of elements in hashset

④ contains() hs.contains()
returns true if it is present, else false

HashMap: data structure which contains key-value pairs.



① HashMap can have duplicate values
can't have duplicate keys

② Query on key

③ Pairs are in random order

Suppose we've Indian states and their population

Hashmap

Punjab	: 12
Haryana	: 9
Rajasthan	: 13
UP	: 12

<key, value>

State, population

String, Integer

hm.put("Punjab", 12)

hm.put("Punjab", 20)

① Key has to be unique, so state-name should be key

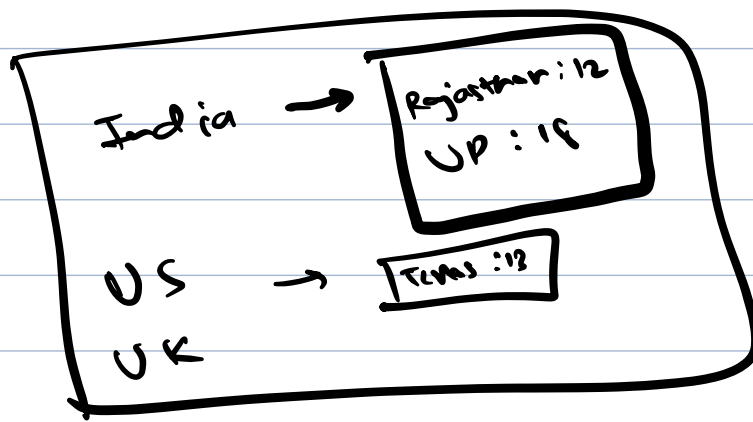
② Do we care about order? X

③ Do we need to search on state key in our data? ✓

Other examples: ① User-id → password

② word → Meaning
(Dictionary)

Country → State → Pop



Syntax:

```
HashMap <Key Type, Value Type> hm =  
    new HashMap <Key Type, Value Type>();
```

Methods

- ① put (x, y) : Add x as key and y as value in hashmap
- ② containsKey (x) : check if a key 'x' is present or not
- ③ remove (x) : remove the key x and its value from hashmap
- ④ size ()
- ⑤ get (x) : given the key x is present it returns associated value

Q.

< 1, 4, 3, -2, 1, 1, 4, 5, 3 >

elem: frequency

1	:	X	3
4	:	X	2
3	:	X	2
-2	:		1
5	:		1

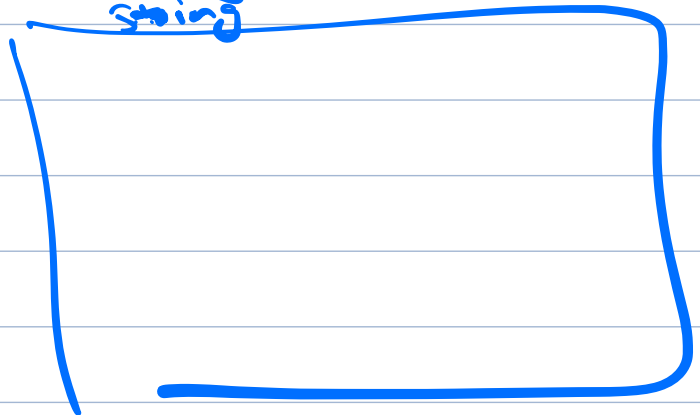
HM

Suresh →

person, city — Designation

key string → value

ram - delhi -



① 1p. and 1m → 2

② 1a and 1m → 2

~~m~~~~p~~~~a~~~~m~~~~p~~~~a~~~~m~~~~m~~

"m,p,a"

cnt a's , p's and m's
↓ ↓ ↓
x + y = z

if (cnt of a's + cnt of p's =
cnt of m's)

return 1